Wetland Segmentation Using Sentinel-2 Imagery and U-Net with Explainability Analysis

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Introduction

- Wetland segmentation using satellite imagery has been widely studied due to ecological significance.
- U-Net is still widely used due to its simple architecture among recent models [1].
- Deep learning models are often difficult to interpret [2].
- This project evaluates U-Net performance and model explainability for wetlands in Ontario.

Datasets

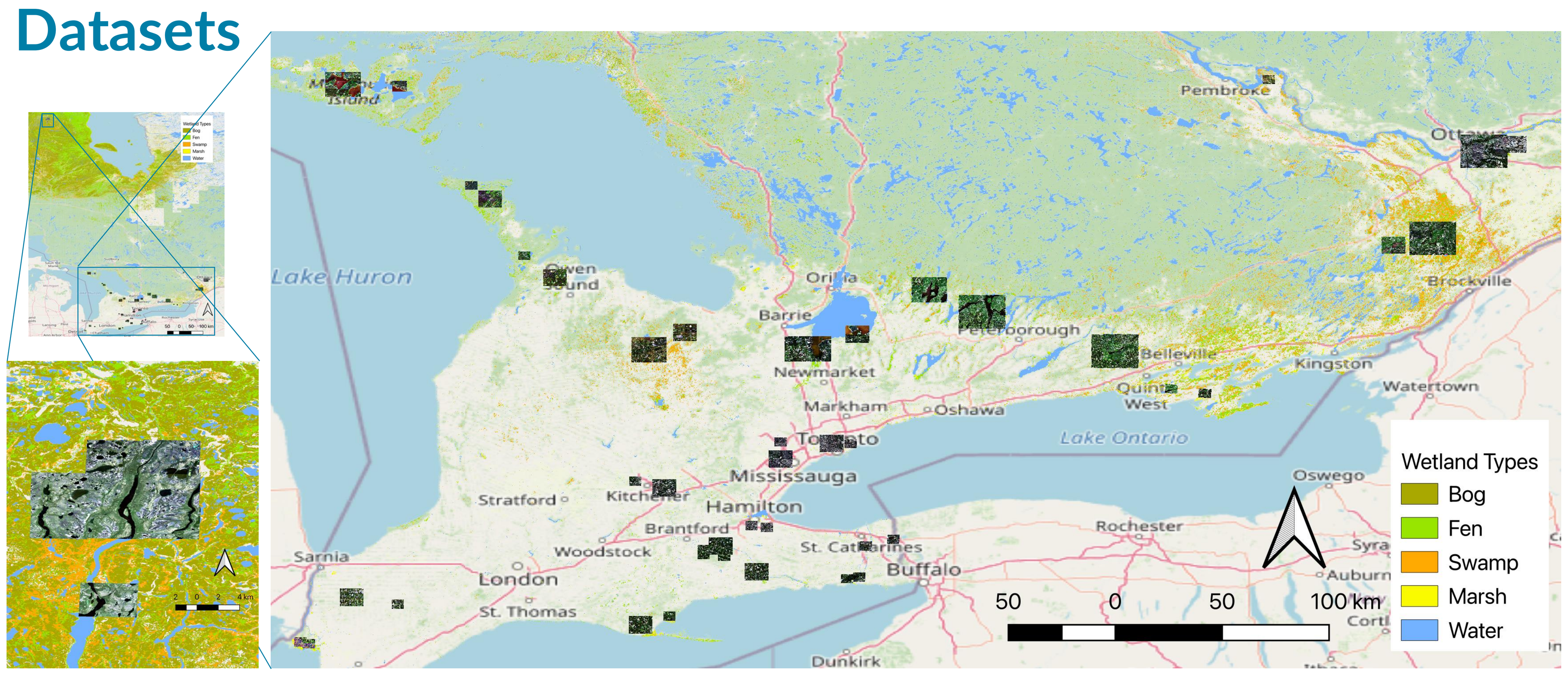


Fig. 1. Wetland types in Ontario and study areas.

Satellite Imagery

- Sentinel-2 Level 2A
- Bands: 2, 3, 4, 8, 11, 12
- Cloud cover: < 20%
- Dates: Jun. 1, 2021 – Sep. 30, 2021
- Study areas: Multiple square patches (256 x 256 pixels)

Ground Truth

- Canadian Wetland Inventory Map Version 3A (CWIM3A, released in 2022)

Total Samples

- **Train: 580 patches, Test: 186 patches**

Methodology

Main Model: U-Net

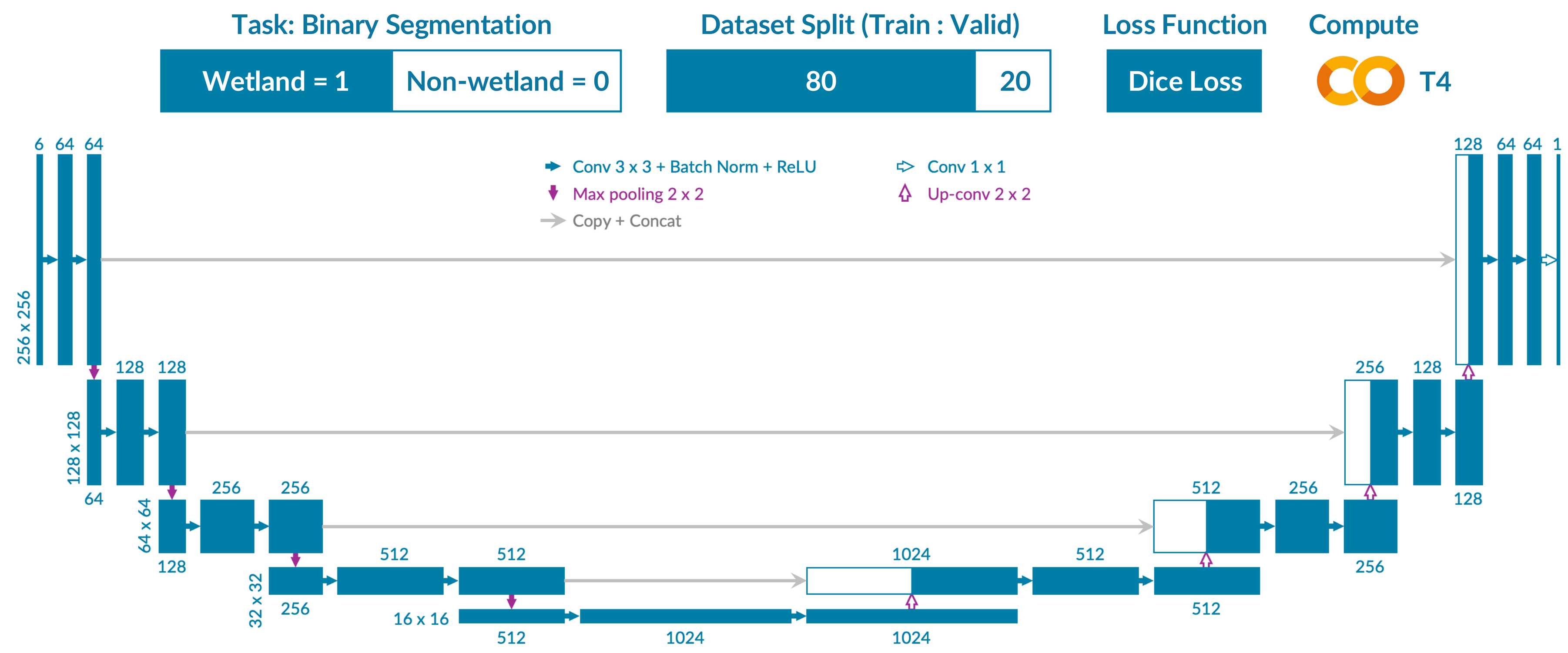
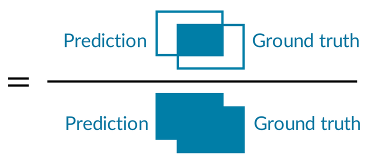


Fig. 2. The U-Net architecture used in this project.

Baseline Model: Random Forest

Evaluation Metric: Intersection over Union (IoU*)

$* \, IoU =$



Explainability Analysis

(1) Wetland type-wise: Coverage rate and FPR for each wetland class

$$Coverage_c = \frac{TP_c}{GT_c} \qquad FPR = \frac{FP}{Non\ wetland}$$

TP_c = pixels predicted as wetland AND class is c
GT_c = total pixels of class c
FPR = false positive rate
FP = false positive

(2) Band-wise: Band importance using band occlusion

(3) Geometry-wise: Grad-CAM

Results

Model Comparison

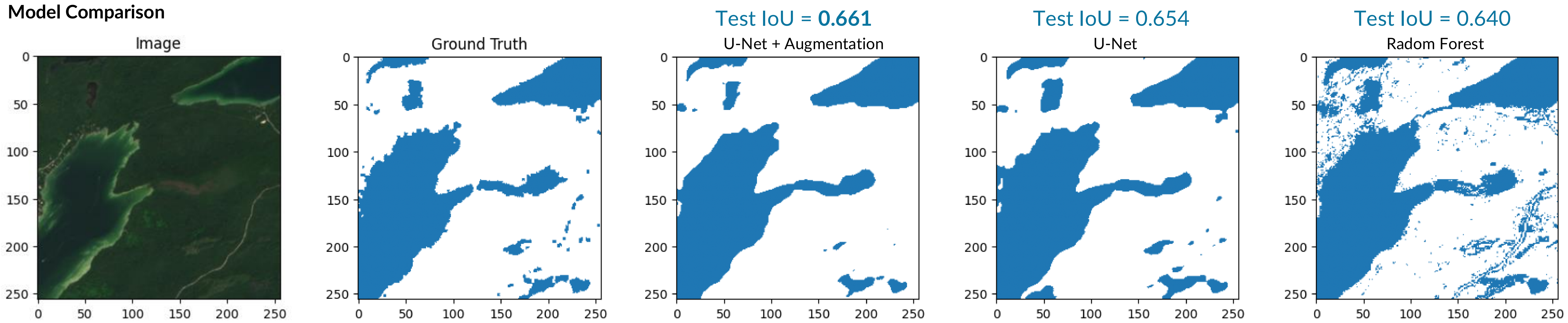


Fig. 3. Segmentation results for a study area in Bruce Peninsula National Park using three models.

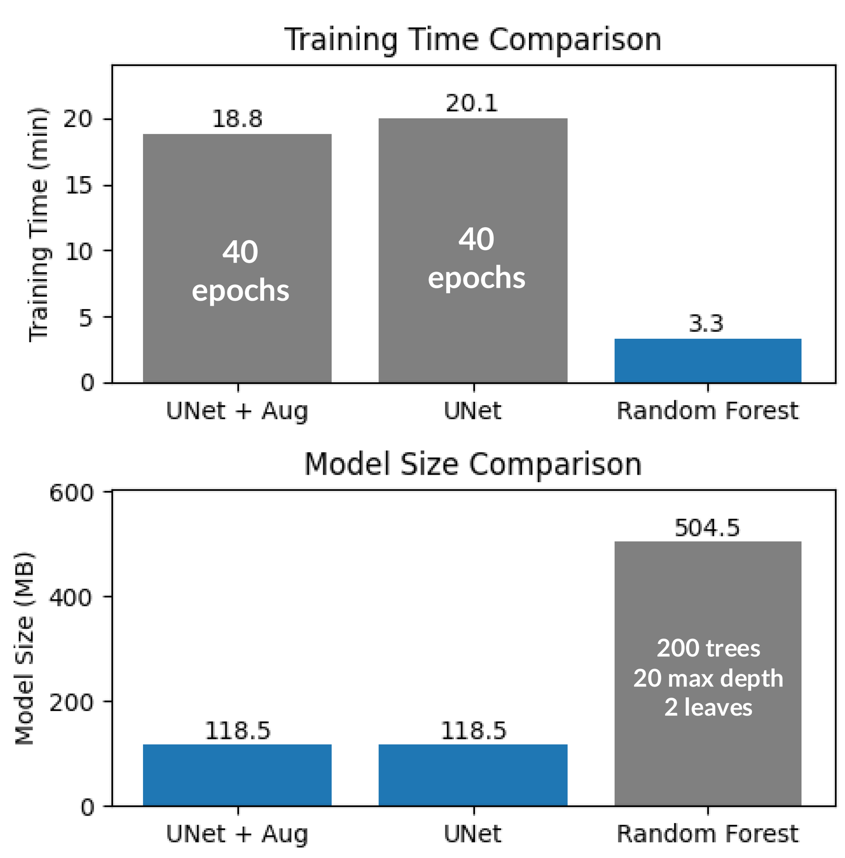


Fig. 4. Model comparison: training time and model size.

Explainability Analysis

(1) Wetland type-wise

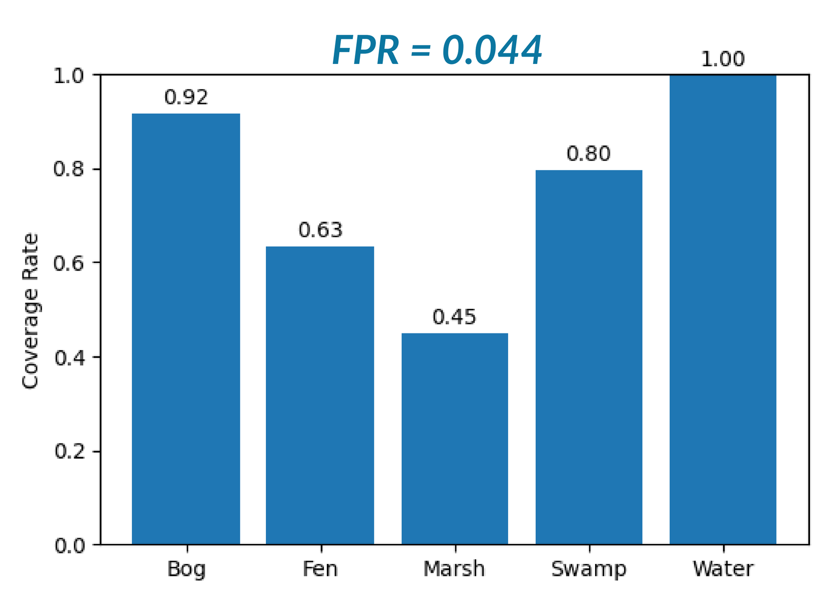


Fig. 5. Coverage rate by wetland type.

(2) Band-wise

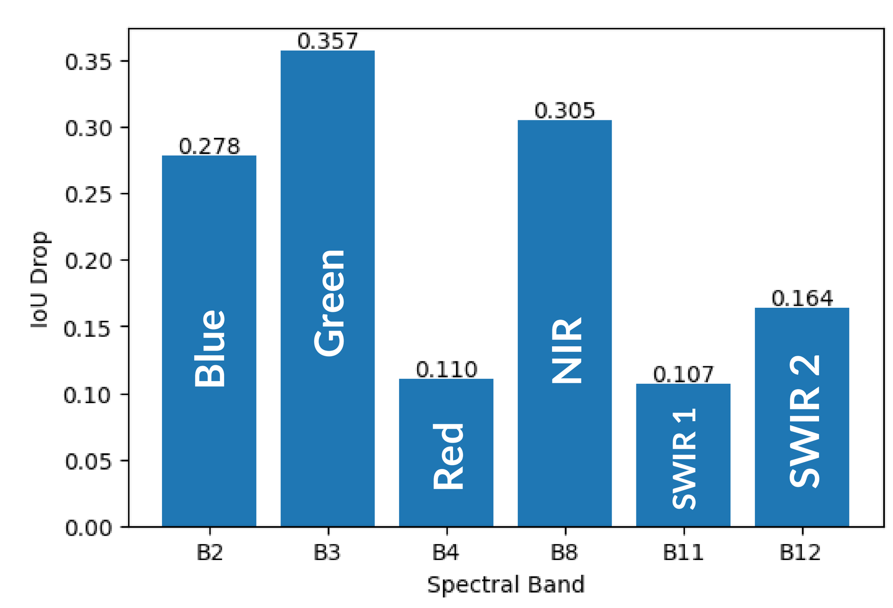


Fig. 6. Band importance from band occlusion analysis.

(3) Geometry-wise

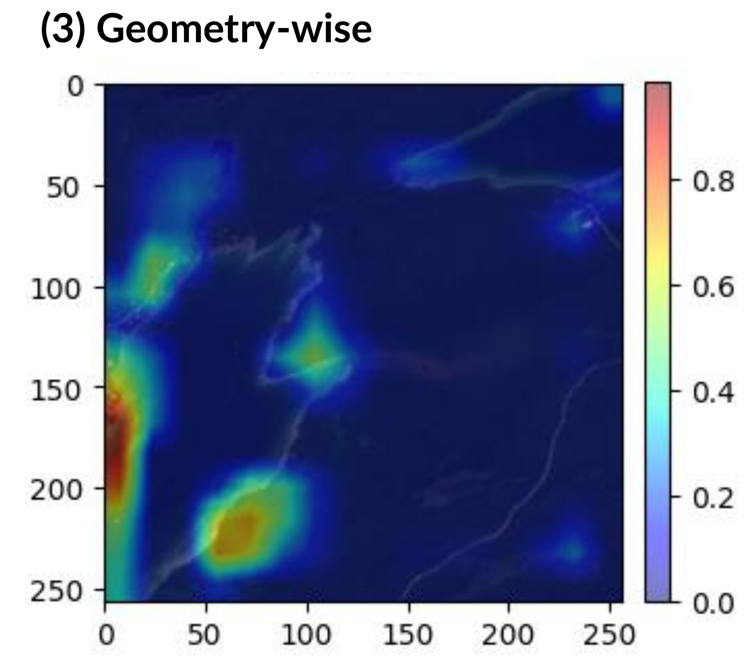


Fig. 7. Grad-CAM highlights for model interpretation.

Discussion

- **U-Net is effective due to smaller model size and better accuracy.**
- **Augmentation might not be necessary when enough data are available.**
- **Water areas tend to be detected more easily than other types.**
- **B2 (Blue), B3 (Green), and B8 (NIR) contribute more among bands.**
- **U-Net reacts to wetland boundaries and mixed areas.**
- **Model performance on newer images (e.g., 2025) or other seasons was not evaluated.**
- **Strict input data preprocessing (e.g., size: 256 × 256 pixels) is required**
- **Multiclass segmentation might improve functionality and potentially accuracy.**

References

[1] Merchant, M. A., Mahdianpari, M., Bourgeau-Chavez, L., DeVries, B., & Berg, A. (2025). Automated Semantic Segmentation of Arctic Surface Water Features with Very-High Resolution Satellite X-Band Radar Imagery and U-Net Deep Learning: Segmentation sémantique automatisée des caractéristiques des eaux de surface de l'Arctique à partir d'images radar satellite en bande X à très haute résolution et à l'aide de l'apprentissage profond U-Net. Canadian Journal of Remote Sensing, 51(1), 2533460.

[2] Rony, J. H., Al Shafi, A., Khan, R. H., Syeed, M. M., & Uddin, M. F. (2025, November). Semantic Segmentation in Natural Environments: An Analysis of Deep Learning Techniques. In 2025 IEEE International Performance, Computing, and Communications Conference (IPCCC) (pp. 1-6). IEEE.